

1. Administrative & Study Identification

Full Study Title: A Phase II Multicenter, Open-label, Single-arm Dose Escalation Study of Asciminib Monotherapy in 2nd and 1st Line Chronic Phase - Chronic Myelogenous Leukemia (ASC2ESCALATE)
Brief Title: Asciminib Monotherapy, With Dose Escalation, for 2nd and 1st Line Chronic Myelogenous Leukemia **Short Title/Acronym:** ASC2ESCALATE **Protocol Number:** CABL001AUS08 **NCT Number:** NCT05384587 **Posting ID:** 690979451729129044 **Trial Status:** Active, Not Recruiting (ACTIVE_NOT_RECRUITING) **Sponsor/Company:** Novartis **Investigational Product:** Asciminib (Scemblix®) **ATC Code:** L01EA06 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/scemblix-epar-product-information_en.pdf **Phase of Development:** PHASE2 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy—specifically the Major Molecular Response (MMR) rate at 12 months—of asciminib monotherapy using a dose-escalation strategy in patients with chronic myeloid leukemia in chronic phase (CML-CP) who have been previously treated with one prior ATP-binding site TKI (2L cohort). **Secondary Objectives:** To evaluate the MMR rate at 12 months in the newly diagnosed 1L cohort; to assess MMR and MR2 rates at various visits; to evaluate rates of deep molecular response (MR4.5) at 24 and 36 months; and to monitor safety, tolerability, progression-free survival (PFS), and overall survival (OS).

2.2 Background & Rationale ATP-competitive tyrosine kinase inhibitors (TKIs) have drastically extended life expectancy in CML. However, up to 40% of newly diagnosed patients with CML-CP stop first-line therapy within 5 years due to resistance or intolerance. Asciminib is a first-in-class STAMP (Specifically Targeting the ABL Myristoyl Pocket) inhibitor that maintains activity against most BCR::ABL1 mutations. This trial aims to address the unmet need in the 2nd-line (after 1 prior TKI) and 1st-line settings by utilizing a dose-escalation strategy for patients not meeting strict molecular milestones, aiming to maximize deep responses and prevent disease progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Single-arm

assignment to either the 1L or 2L cohort based on prior treatment history)

3.2 Planned Enrollment & Duration Targeted Enrollment: 34 participants as per primary registry parameters (with broader cohort enrollment planning of 92 patients for the 2L cohort, and between 60 to a maximum of 90 patients for the 1L cohort). **Number of Sites:** Multicenter (~40 sites across the United States) **Estimated Study Start:** August 2022 **Estimated Primary Completion:** October 2027

4. Subject Selection (Eligibility Criteria)

4.1 Key Inclusion Criteria 1. Age ≥ 18 years with an ECOG performance status of 0, 1, or 2. 2. Documented diagnosis of CML-CP, with no previous history of accelerated phase (AP) or blast crisis (BC). 3. For 2L Cohort: Discontinuation of 1 prior ATP-binding site TKI due to treatment failure, warning (according to 2020 ELN Recommendations; Hochhaus et al), or intolerance. For 1L Cohort: Newly diagnosed CML-CP (up to 4 weeks of prior TKI allowed). 4. Adequate end organ function within 14 days prior to the first dose of asciminib, defined as: * Total bilirubin $\leq 3.0 \times \text{ULN}$ without AST/ALT increase * AST and ALT $\leq 5.0 \times \text{ULN}$ * Alkaline phosphatase $\leq 2.5 \times \text{ULN}$ * Serum lipase $\leq 1.5 \times \text{ULN}$ (values $> \text{ULN}$ and $\leq 1.5 \times \text{ULN}$ must not be clinically significant or associated with acute pancreatitis risk factors) * Creatinine clearance ≥ 30 mL/min (Cockcroft-Gault formula)

4.2 Key Exclusion Criteria 1. Previous treatment with 2 or more ATP-binding site TKIs (for 2L) or > 4 weeks with 1 TKI (for 1L). 2. Previous treatment with asciminib or known presence of the T315I mutation at any time prior to study entry. 3. Known second chronic phase of CML after previous progression to AP/BC, or previous/planned hematopoietic stem-cell transplantation. 4. History of myocardial infarction, angina pectoris, or coronary artery bypass graft (CABG) within 6 months prior to starting study treatment, or clinically significant cardiac arrhythmias. 5. QTcF ≥ 450 msec (male and female) at screening, or history of Long QT syndrome/risk factors for Torsades de Pointes. 6. History of acute pancreatitis within 1 year of study entry or past medical history of chronic pancreatitis. 7. Pregnant or nursing (lactating) women; women of child-bearing potential or sexually active males not using highly effective contraception.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Initial dose of asciminib is 80 mg once daily (QD). A protocol-defined dose escalation occurs if molecular milestones are not met: at 6 months, patients not achieving $\text{BCR}::\text{ABL1}^{\text{IS}} \leq 1\%$ escalate

to 200 mg QD. At 12 months, patients not achieving MMR escalate to 200 mg QD (if on 80 mg) or 200 mg twice daily (BID) (if already on 200 mg QD). **Route of Administration:** Oral **Duration of Treatment:** Active treatment for up to 156 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for concurrent conditions.
Prohibited: Concurrent treatment with other TKIs or investigational antineoplastic agents during the active treatment phase. Medications that are strong inducers of CYP3A (for patients on 80 mg QD and 200 mg QD) or strong inducers and inhibitors of CYP3A (for patients on 200 mg BID) must be switched to alternatives at least one week prior to treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, T315I mutation testing, Baseline labs/ECG
Baseline	Day 0	Enrolment, First Dose of Asciminib (80 mg QD)
Interim 1	Month 6	Efficacy Assessment (BCR::ABL1 PCR). Dose escalation to 200 mg QD if $BCR::ABL1^{IS} > 1\%$
Interim 2	Month 12	Primary Efficacy Assessment (MMR). Dose escalation to 200 mg QD or 200 mg BID if MMR not achieved
End of Study	Week 156	Final Efficacy/Safety Assessments, 30-day Safety Follow-up period

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Major Molecular Response (MMR) rate at 12 months (48 weeks) in the 2L patient cohort.
Measurement Method: $BCR::ABL1^{IS} \leq 0.1\%$ measured by real-time quantitative Polymerase Chain Reaction (RQ-PCR).

7.2 Secondary & Exploratory Endpoints

- Percentage of participants achieving Molecular Response 4.5 (MR4.5) at 24 and 36 months.
- MMR Rate at visits other than the 12-month mark.
- MR2 Rate at visit.
- Progression-free survival (PFS) and overall survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection of clinical and laboratory AEs (including monitoring for pancreatic toxicity via serum lipase, cardiovascular events, and hypertension). **Serious Adverse Events (SAEs):** Reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring conducted by the sponsor to assess dose-escalation safety and general tolerability profiles.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Full Analysis Set (all enrolled patients who receive at least one dose of study medication) for efficacy evaluations. Safety Set for all AE/SAE analyses. **Handling of Missing Data:** Patients who discontinue treatment, lose response, or are missing assessment data at the designated milestone (e.g., 12 months) are considered non-responders for that specific time point.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to verify informed consent, eligibility, source data (specifically PCR laboratory reports), and safety reporting. **Audit Trail:** Secure eCRF locking, query resolution processes, and data clarification forms via standard Novartis QA audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CABL001AUS08 **Registry Number:** NCT05384587 **Posting ID:** 690979451729129044 **Sponsor/Lead Organization:** Novartis **Study Title:** ASC2ESCALATE **Official Regulatory Title:** A Phase II Multicenter, Open-label, Single-arm Dose Escalation Study of Asciminib Monotherapy in 2nd and 1st Line Chronic Phase - Chronic Myelogenous Leukemia

12. Clinical Framework (CML Specific)

Primary Condition / Disease Area: Chronic Myelogenous Leukemia - Chronic Phase (CML-CP) **Preferred UMLS Name:** Chronic Myelogenous Leukemia, Chronic Phase **MeSH Code:** D015466 **SNOMED ID:** 413847001 **Diagnosis Threshold:** Presence of Philadelphia chromosome / \$BCR::ABL1\$ positive without T315I mutation **Medical Methodology:** Tyrosine Kinase Inhibitor (TKI) Therapy (STAMP inhibition) **Optimization Target:** Major Molecular Response (MMR; \$BCR::ABL1^{IS} \leq 0.1\%\$)

13. Study Procedures & Methodology

Management Pathway: TKI dose-escalation step-up pathway **Education Protocol (HCP):** Protocol-specific training on \$BCR::ABL1\$ milestone tracking and dose-escalation criteria (6 and 12 months) **Education Protocol (Patient):** Oral chemotherapy adherence, self-monitoring, and scheduled molecular testing awareness **Quality Audit Mechanism:** Routine central/local laboratory \$BCR::ABL1\$ PCR

tracking and cardiovascular/pancreatic laboratory safety monitoring

Total Duration: Up to 156 weeks of active treatment + 30-day safety follow-up **Onsite Visit Cadence:** Baseline, and regularly scheduled interval visits primarily aligned with molecular milestone testing (Weeks 12, 24, 48, 72, 96, up to 156) **Remote Monitoring Frequency:** Standard risk-based monitoring site contacts **Checklist Review Interval:** Routine molecular response checks specifically at 3, 6, 12, 18, 24, and 36 months

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					